

# Snow impact on PV performance: Assessing the zero-output challenge in cold areas

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## ABSTRACT

Solar photovoltaic (PV) technology has a great potential for renewable energy generation. However, in cold climates with heavy snowfall, PV systems performance might be significantly reduced. This review investigates the impact of snow on solar PV in regions with harsh winters. It describes the snow soiling process while determining the main factors leading to snow losses in PV plants. Factors influencing snow losses are classified into four major categories: controllable factors, such as surface properties and PV installation characteristics, and uncontrollable factors, including climatic conditions and snow properties. The literature review reveals significant variations in reported snow losses due to the number of influential factors. One key recommendation is to improve PV system design to better accommodate snowy conditions, rather than relying on configurations optimized for milder conditions. The review also identifies a gap in the literature regarding the implementation of safety devices such as snow guards in the context of PV systems. With the increasing adoption of bifacial technologies, further research is needed to assess the benefits of snow's high albedo. For accurate snow loss modeling and shedding prediction, it is crucial to include additional factors impacting snow accumulation and clearing, such as liquid water content which affects snow adhesion. Additionally, enhancing models through image analysis and integrating drones for precise camera angles can significantly improve snow detection across various PV configurations and locations.

## Nomenclature

| PV    | Photovoltaic                                       |
|-------|----------------------------------------------------|
| AI    | Artificial intelligence                            |
| LWC   | Liquid water content                               |
| PoA   | Plane of Array                                     |
| PV/T  | Photovoltaic-Thermal                               |
| GBT   | Gradient boosted tree                              |
| DL    | Deep Learning                                      |
| LMB   | Levenberg–Marquardt Backpropagation                |
| CNN   | Convolutional neural network                       |
| SAT   | Single axis tracker                                |
| p-Si  | Polycrystalline Silicon                            |
| m-Si  | Monocrystalline Silicon                            |
| CIGS  | Copper Indium Gallium Selenide                     |
| CdTe  | Cadmium Telluride                                  |
| CIS   | Copper Indium Selenide                             |
| BAPV  | Building applied photovoltaic                      |
| NSRDB | National Solar Radiation Database                  |
| TMY2  | Typical Meteorological Year Data of 2nd generation |

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| PV   | Photovoltaic                                    |
|------|-------------------------------------------------|
| TMY3 | Typical Meteorological Year Data 3rd generation |
| EV   | Electric Vehicle                                |
| DC   | Direct Current                                  |
| GCR  | Ground coverage ratio                           |

## 1. Introduction

As the global transition towards renewable energy sources intensifies, understanding the performance of photovoltaic (PV) systems in extreme weather conditions becomes crucial for adapting to changing climate patterns. PV technology faces certain challenges in cold climates such as snow and ice [1] acting as barriers, obstructing light from reaching the cells. In recent years, research on the impact of snow and ice accumulation on PV systems has received attention in many areas

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including the Nordic countries [2]. The authors in this first review on the topic, delves into the challenges associated with snowfall and ice formation on not only PV solar cell roofs but also solar thermal panels and walls. Building upon earlier work by Ref. [3] who pioneered research on snow's effect on PV systems in Canada, this work aims to identify critical aspects related to snow adhesion on PV surface and its impact on overall solar energy efficiency.

It has been found in literature that snow events lead to significant losses in winter period but also opportunities for increased albedo during snowy winters [4]. The work of [5] reported an annual snow loss percentage of 4.5 % for a portfolio in North America. This amounts to about 20 GWh of energy loss or \$5.5 million (USD) in missing incomes for a single year. The work of [6] has estimated PV output losses due to snow cover during the winter period of 2017–2018 by more than 1.5 months of summer production for both locations Helsinki and Kuopio in Finland.

PV performance in snowy conditions has been studied by Refs. [7–9] and for different PV technologies by Ref. [4]. The latter has pointed out that bifacial systems perform better than monofacial in severe winter conditions, where the annual losses for the studied case were 2 % in the worst-case scenario compared to 16 % for monofacial PV. A wide variation in snow losses has been found according to the conducted literature review. Monthly snow losses have reached high values up to 100 % [10–12]. Annual snow losses are of a lower magnitude than monthly losses, ranging from 0.6 % [13] to 20 % [14].

For the aspect of modelling and estimation of energy losses due to snow, several models can provide direct energy loss considering snow effect [15–19]. These methods rely on data-driven systems and meteorological parameters to predict loss directly. Others can be categorized as snow cover prediction models that estimate the snow-covered area of a panel and simulate the clearing process followed by the calculation of generation loss [11,20–22]. Threshold-type and first principle thermal heat and mass transfer models have been used to assess snow coverage on PV panels as reported by Ref. [23]. These models rely on several factors related to climatic conditions such as snow depths and snowfall amounts and PV installation characteristics as tilt angle [15,17,19,21,24]. A recent work of [22] has incorporated more factors such as ambient and module temperature, wind speed and wind direction, humidity and air pressure. Given the complexity of the topic, some works have attempted to utilize machine learning techniques to formulate a prediction problem [25].

Snow mitigation and shedding approaches were among the reported topics in previous reviews [2,23,26–29]. Active solutions include heating and mechanical shedding of snow while passive solutions involve the application of coating on PV surface and optimize design for increased natural shedding depending on weather conditions. Based on the advantages and limitations of these solutions, it has been recommended to combine heating and surface coatings to increase the sliding of snow [30].

This work reviews for the first time the full process of snow soiling and associated energy losses, detailing the transport, deposition, and accumulation stages. The impact on PV performance is described by highlighting the challenges faced by snow accumulation for various PV technologies and climate zones according to Köppen Geiger classification, and by providing an overview of reported snow loss values worldwide. A comprehensive analysis is given of the factors influencing snow soiling, classified into controllable and uncontrollable categories, and of the models used to estimate and predict snow losses with an emphasis on their limitations and the most important factors to be incorporated. Finally, a summary is given of the existing solutions for snow mitigation and shedding, while underlining new combinations between passive, active and natural snow clearance.

## 2. Meta-analysis

Based on the keywords snow combined with PV performance and

solar panels, including the variants photovoltaic and solar modules, the research for relevant publications has been carried out without time restrictions in the following databases: Scopus, ScienceDirect, Google Scholar, Wiley, Springer and Taylor & Francis. These scientific publications comprise mainly research articles, conference papers and reviews. The articles are summarized in Table 1 and have been distributed according to the location of the study, while the reviews have been classified in Table 2 according to their main studied aspect. Table 1 specifies if the study focuses on snow impact, or its mitigation as illustrated in Fig. 1. Around 44 papers have studied the impact of snow either through field measurements, modelling or simulation while 19 papers have addressed snow mitigation solutions. Besides Canada and United States where most studies have been conducted, attention regarding the snow impact on PV performance has been recently received in Nordic countries followed by East Asia and central Europe.

According to the summary of publications in Table 1, the distribution of published works by PV technology has been also determined. In fact, more than 40 papers studied systems with monofacial PV modules compared to only 7 papers that studied bifacial technology. It should be noted that the remaining works did not define nor mention any indication regarding the PV technology. Relying on the reported recommendations by 43 papers, Fig. 2 shows the distribution of published works. Indeed, 49 % of papers have recommended the optimization of PV design considering several parameters as tilt angle [31], gap to ground distance [32], opting for a specific PV technology as bifacial PV [4] and adjusting the modules frame structure [33,34]. In the work of [35], the panel orientation has been investigated to highlight the potential benefits of portrait versus landscape configuration for snow mitigation. The second major category corresponds to the enhancement of modelling of the losses in energy yield due to snow or their forecasting [19,25], while the prediction of snow detachment and sliding was reported by Ref. [36]. The last category concerns snow mitigation approaches with 12 % of papers that have recommended active solutions such as heating [37,38], passive solutions for snow mitigation [39] and the combination of both [30]. In this work, each of these sections will be detailed while discussing their advantages as well as their limitations.

According to the reviews gathered on the impact of snow on PV systems, further detailed in Table 2 and it has been found that half of the reviews focus on snow removal approaches. Following this, 22 % of the works have highlighted the primary factors influencing snow losses with an additional 14 % discussing snow formation. The remaining 14 % of studies are evenly distributed between forecasting snow events and estimating snow losses.

## 3. Snow soiling process

This section provides an overview of the existing literature on the physics of the snow soiling process, specifically focusing on the transport, deposition, and accumulation of snow particles. It is important to note that none of the previous studies have delved into the snow soiling process in relation to photovoltaic systems.

### 3.1. Transport

In addition to snowfall, snow particles can be redistributed on the ground. Both during snowfall and in the case of horizontal redistribution of snow on the ground, wind impacts transport of snow particles. Horizontal transport of snow particles is classified into three modes: creep, saltation and suspension depending on height as illustrated in Fig. 3 [74]. The friction velocity, measured in meters per second, is crucial in determining the force exerted on particles at ground level. While primarily associated with particle transport, friction velocity also significantly influences the deposition and accumulation of air-blown snow particles [8].

**Table 1**

Summary of published papers related to snow. Category of the papers is determined based on the main scope of the work. Rec. stands for recommendations. \* is associated to works if indication of the manufacturer or the reference of the PV module without directly mentioning the word “Monofacial”. N/A is used when technology is not defined.

| Ref. | Location                                                       | Tilt angle                                           | Technology  | Main findings                                                                                                                                                                                                                                                                                                                                                 | Category   | Rec.                                                                                    |
|------|----------------------------------------------------------------|------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------------|
| [40] | Norway<br><i>Oslo</i>                                          | 10°                                                  | Monofacial  | Tests shown that active snow melting effectively removes snow, but cold and windy conditions are unfavorable for melting.                                                                                                                                                                                                                                     | Impact     |                                                                                         |
| [35] | USA<br><i>Vermont</i>                                          | 30°                                                  | Monofacial  | Image analysis indicates that modules in portrait orientation yield up to 24 % more energy yield compared to landscape orientation during a single snow shedding event. Snow mitigation can be increased by considering the system design.                                                                                                                    | Impact     | Design:<br>Portrait module orientation                                                  |
| [41] | Canada<br><i>North Cape</i>                                    | 30°                                                  | Monofacial  | An artificial intelligence (AI) algorithm is used to process images of snow coverage on a fixed tilt angle array in Canada. The presented models can be improved by using site specific model parameters.                                                                                                                                                     | Impact     | Models:<br>Improved snow loss models                                                    |
| [30] | USA<br><i>Toledo</i>                                           | 25°                                                  | Monofacial  | Different mechanisms of natural snow removal were categorized using a novel sensor that measures the LWC and temperature of the snow at its interface with the modules. The paper presents the impact of snow thickness as well and the LWC on occurrence and duration of each mechanism.                                                                     | Mitigation | Mitigation: Combination of active (heating) and passive (coating) snow removal          |
| [33] | Canada<br><i>Kingston</i>                                      | 30°<br>45°<br>55°                                    | Monofacial  | The authors investigated two heating methods, including electrical heating by a resistance thin-film heater mounted on the back of the PV modules and by the reverse current through PV cells. It was observed that the entire PV surface requires heat to remove snow, as the panel thermal conduction was not sufficient to conduct heat to unheated areas. | Mitigation | Design:<br>Frameless structure                                                          |
| [42] | Finland<br><i>Oulu</i>                                         | 90°<br>23°<br>28°<br>37°<br>41°<br>42°<br>46°        | Monofacial  | This paper presents the impact of snow on PV in Finland. It was found that lower tilt angles proven to be the most effective, capturing the spring and summer irradiation.                                                                                                                                                                                    | Impact     | Design:<br>Optimal tilt angle                                                           |
| [43] | USA                                                            | N/A                                                  | N/A         | Operation and maintenance tickets, weather and production data were implemented in statistical techniques and machine learning to evaluate performance impacts in difficult weather conditions. Snow events cause the greatest reduction in performance of 54.5 %.                                                                                            | Impact     |                                                                                         |
| [44] | Canada<br><i>Edmonton</i>                                      | 21°<br>27°<br>30°<br>31°<br>33°<br>45°<br>47°<br>71° | Monofacial  | Sensitivity analysis that examines the effectiveness of absorbed, accumulated solar irradiances and thermal capacitance of PV on the prediction process of snow cover was conducted. Using absorbed irradiance significantly improves the prediction model compared to those based on plane of array irradiance (PoA).                                        | Impact     | Models:<br>Use of absorbed irradiance rather than PoA                                   |
| [45] | China<br><i>Xinjiang</i>                                       | 30°                                                  | Monofacial* | Regarding snow, it has been found that the relative humidity is rapidly decreased when snow covered the ground.                                                                                                                                                                                                                                               | Impact     | Models:<br>PV climate model                                                             |
| [46] | Canada<br><i>Kingston</i>                                      | 45°                                                  | Monofacial  | Circulating hot fluid through the back of the panel has been proposed for removing snow from PV/T panels. This method has shown its effectiveness in snow removal. Measurements have been compared to those given by numerical model.                                                                                                                         | Mitigation | Design:<br>Frameless structure and better design for the rear surface of the panel      |
| [47] | Canada<br><i>Ontario</i>                                       | 5°<br>10°<br>15°<br>20°<br>40°<br>60°                | Monofacial  | Performance data extracted have been used to validate a novel methodology to determine the snowfall losses based on image analysis. It was found that high tilt angles can increase PV yield considering the effect of increasing the albedo.                                                                                                                 | Impact     | Design:<br>Better exploitation of the albedo<br>Proper characterization of the snowfall |
| [39] | Canada<br><i>Kingston</i>                                      | 10°<br>20°<br>40°<br>60°                             | Monofacial  | This study presents a new methodology to determine the effectiveness of snow clearing from PV panels based on hydrodynamic coatings. The tested coatings did not have appreciable benefits.                                                                                                                                                                   | Mitigation | Mitigation: Coatings enhancement                                                        |
| [11] | USA<br><i>Colorado</i><br><i>Wisconsin</i>                     | 15°<br>18.4°<br>22.5°<br>26°<br>30°<br>35°           | Monofacial  | Energy production losses due to snow were measured in Colorado and Wisconsin locations. A prediction model of performance losses caused by snowfall is also developed. Losses in annual energy production were ranged from 1 % to 12 %.                                                                                                                       | Impact     |                                                                                         |
| [12] | Norway                                                         | 10°<br>24°<br>26°<br>40°                             | Monofacial  | In this work, a snow depth dependent clearing rate coefficient in the Marion snow loss model has been presented considering the effect of different snow conditions on the natural snow clearing process.                                                                                                                                                     | Impact     |                                                                                         |
| [48] | Canada<br><i>Edmonton</i><br><i>Calgary</i><br><i>Red Deer</i> | 9°<br>14°<br>18°<br>27°<br>30°<br>34°                | N/A         | This paper develops a predictive model for daily PV system generation using long-term data from 85 sites. Snow coverage loss factors are empirically derived to enhance model performance. An artificial neural network model is then implemented, considering snowfall data and snow coverage loss factors.                                                  | Impact     | Design:<br>Higher tilt angles                                                           |

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Table 1 (continued)

| Ref. | Location                                     | Tilt angle                      | Technology             | Main findings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Category   | Rec.                                                                              |
|------|----------------------------------------------|---------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------|
|      |                                              | 37°<br>40°<br>60°<br>26.56°     |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |                                                                                   |
| [24] | USA<br><i>Green Bay</i>                      |                                 | Monofacial*            | The aim of this paper is to demonstrate the need to modify the current solar energy performance models to include snow. A new model has been introduced to decrease forecasting error.                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Impact     | Models:<br>Model enhancement                                                      |
| [49] | Germany<br><i>Darmstadt</i>                  | 15°<br>30°                      | N/A                    | The authors have investigated the snow accumulation patterns on flat roofs with solar arrays installed at small tilt angles. A water channel using sand was chosen to imitate snow particles.                                                                                                                                                                                                                                                                                                                                                                                                                                                | Impact     | Design:<br>Providing space for snow accumulation<br>Local snow load investigation |
| [50] | Norway                                       | 10°                             | N/A                    | Performance metrics have been tested on PV installations namely specific yield, temperature corrected performance ratio and power performance index based on both physical modelling and machine learning. According to this paper, snow is one among the most influential effects leading to instability in the performance metrics. The results of this study found that monofacial snow losses are in average 33 % for winter period, and 16 % on an annual basis. Bifacial PV performs better than monofacial in severe winter conditions as annual snow losses was 16 % for monofacial and 2 % for bifacial in the worst-case scenario. | Impact     | Models:<br>Data quality and selection of suitable performance metric              |
| [4]  | USA<br><i>Escanaba</i>                       | 35°                             | Monofacial<br>Bifacial | The results of this study found that monofacial snow losses are in average 33 % for winter period, and 16 % on an annual basis. Bifacial PV performs better than monofacial in severe winter conditions as annual snow losses was 16 % for monofacial and 2 % for bifacial in the worst-case scenario.                                                                                                                                                                                                                                                                                                                                       | Impact     | Design: Optimization of interrow spacing involving bifacial PV                    |
| [51] | Canada<br><i>Ontario</i>                     | 30°                             | Monofacial*            | In this work a PV modelling approach has been proposed to represent instantaneous electrical characteristics of PV modules in the presence of snow coverage. The proposed model utilizes the Bouguer–Lambert Law to estimate the level of insolation reaching surface of snow-covered PV cells.                                                                                                                                                                                                                                                                                                                                              | Impact     | Models: Incorporating models in power electronics and control topologies          |
| [19] | USA<br><i>Denver</i>                         | 5°<br>40°                       | Monofacial*            | Snow power loss calculation based on recorded electrical and meteorological data using computational intelligence techniques. Snow loss has been well predicted using gradient boosted tree (GBT).                                                                                                                                                                                                                                                                                                                                                                                                                                           | Impact     | Models:<br>Prediction models                                                      |
| [52] | China<br><i>Las Vegas</i>                    | N/A                             | Monofacial*            | This study presents a method to quantify the snow impact on PV panels using image processing and deep learning (DL) techniques mainly Levenberg–Marquardt Backpropagation (LMB) and Convolutional Neural Network (CNN).                                                                                                                                                                                                                                                                                                                                                                                                                      | Impact     |                                                                                   |
| [53] | Germany<br>Switzerland                       | N/A                             | N/A                    | This work presents a validation study that compares satellite-based snow datasets and meteorological stations data from Germany and Switzerland.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Impact     | Snow detection accuracy                                                           |
| [54] | Germany<br><i>Arzberg</i>                    | 30°                             | Monofacial             | In this paper, the authors analyzed the snow melting behavior on different PV technologies based on image processing. The study has reported high snow removal rates for large frameless modules.                                                                                                                                                                                                                                                                                                                                                                                                                                            | Impact     | Design: Suitable PV technology for snowy area                                     |
| [7]  | USA                                          | N/A                             | N/A                    | Operational PV system data and external snow records have been investigated. It has been found around 2 % snow soiling across the fleet of PV systems in 2018.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Impact     | Models:<br>Irradiance data availability using heated sensors                      |
| [55] | Norway<br><i>Kjeller</i>                     | 10°<br>28°                      | Monofacial*            | Based on data signatures, 97 % of the snow losses in the dataset have been detected. The paper shows that the Marion model outperforms the other models. By improving the Marion model to include system design and snow depth, the modelled absolute error loss is significantly reduced for the tested system, from 23 % to 3 %.                                                                                                                                                                                                                                                                                                           | Impact     | Models:<br>Model improvement                                                      |
| [34] | USA<br><i>Vermont</i>                        | 35°                             | N/A                    | Snow losses can be reduced by the installation of frameless PV modules, or with frames designed to aid snow shedding.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Mitigation | Design:<br>Frameless PV modules                                                   |
| [36] | USA<br><i>Toledo</i>                         | 45°<br>65°                      | N/A                    | It has been found that wind has a significant effect on snow melting. In this study, a wind of an average velocity of approximately 2 m/s at around 3 °C air temperature, has increased snow melting by 90 %.                                                                                                                                                                                                                                                                                                                                                                                                                                | Mitigation | Models:<br>Prediction of snow detachment                                          |
| [56] | Canada<br>Ontario<br><i>Cambridge Bay</i>    | SAT                             | Bifacial               | Modelled rear irradiance has been compared with energy yield for snow-dependent clearance approach against the common fixed one for single-axis tracker (SAT). A daily energy yield loss of 0.034 % per centimeter of accumulated snow was found.                                                                                                                                                                                                                                                                                                                                                                                            | Impact     | Models:<br>Real time and short time forecasting                                   |
| [9]  | Norway                                       | 10°<br>24°<br>26°<br>40°        | Monofacial*            | Snow losses have been estimated from historical data for a set of PV plants in Norway. The study has described the variations in both monthly and annual snow losses to be included in e.g. PV yield modelling.                                                                                                                                                                                                                                                                                                                                                                                                                              | Impact     | Models:<br>Include snow in PV yield modelling                                     |
| [8]  | Norway<br>Svalbard<br><i>Queen Maud Land</i> | 10°<br>30°<br>50°<br>70°<br>90° | Monofacial<br>Bifacial | The paper demonstrates that increasing the gap-to-ground and the system scale are the only parameter adjustments that reduce the risk of snowdrift accumulation while having an insignificant impact on energy yield.                                                                                                                                                                                                                                                                                                                                                                                                                        | Impact     | Design:<br>Climate-dependent design                                               |
| [57] | Norway<br>Svalbard                           | 30°                             | Monofacial<br>Bifacial | The study shows that snowdrifts pose a challenge, but the low temperatures in Polar climates enhance module performance, and an adaptation of solar power plant design is necessary to mitigate snowdrift development. Snow cover in Polar climates leads to a strong seasonal variation in bifacial gain, which averages 14.7 % annually.                                                                                                                                                                                                                                                                                                   | Impact     | Design:<br>Climate-dependent design                                               |
| [6]  | Finland<br><i>Helsinki</i><br><i>Kuopio</i>  | 15°                             | Monofacial             | This study presents findings regarding the performance of a commonly used parametric PV output model in Nordic conditions. It estimates that PV output losses due to snow cover during the winter period of 2017–2018 could extend to up to 1.5 months of summer production.                                                                                                                                                                                                                                                                                                                                                                 | Impact     |                                                                                   |

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Table 1 (continued)

| Ref. | Location                                                              | Tilt angle                                         | Technology             | Main findings                                                                                                                                                                                                                                                                                                     | Category   | Rec.                                                 |
|------|-----------------------------------------------------------------------|----------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------|
| [58] | Canada<br>Kingston                                                    | 20°<br>45°<br>60°                                  | Monofacial             | This work presents some numerical simulations as well as a parametric study of wind speed, tilt angle and surface emissivity of the insulation on the panel temperature. Using this method to improve snow removal is useful in regions with energy loss of 5 % or more due to snow.                              | Mitigation | Design:<br>Insulation on the back of PV panel        |
| [31] | Sweden<br>Hammarby<br>Sjöstad                                         | 15°<br>40.7°                                       | Monofacial             | It has been found that optimal installation angle could yield approximately a 4.8 % improvement in the studied case compared to traditional angles. The study also explores the application of snow removal solutions which could potentially increase PV generation by 0.1–2.3 %.                                | Mitigation | Design:<br>Optimal tilt angles                       |
| [59] | China<br>Shanghai                                                     | 18°                                                | Monofacial*            | Self-heating can effectively remove snow from PV panels when the snow thickness exceeds the equivalent height and the PV tilt angle is greater than the required minimum, as determined by the mechanical model                                                                                                   | Mitigation | Design:<br>Higher tilt angles                        |
| [60] | USA<br>Calumet                                                        | 0°<br>15°<br>30°<br>45°                            | Monofacial*            | Annual energy losses due to snow are significantly dependent on the tilt angle and the degree of ground interference. It was found that higher tilt angle of around 45° combined with snow removal geometries are recommended to maintain the initial energy production.                                          | Impact     | Design:<br>Higher tilt angles                        |
| [17] | Canada<br>Kingston                                                    | 10°<br>20°<br>40°<br>70°                           | Monofacial*            | This work presents a method to identify snowfall effects from collected performance data. This latter recommends a model to predict these effects based only on meteorological data. This study has shown that low tilt angles are most affected.                                                                 | Impact     | Models:<br>Snow losses modelling                     |
| [61] | USA<br>Truckee                                                        | 0°<br>24°<br>35°<br>39°                            | Monofacial             | This paper introduces a snow loss model that has been calibrated using data from PV systems installed at different tilt angles. The study highlights the significant effect of ground interference on snow.                                                                                                       | Impact     |                                                      |
| [62] | Turkey                                                                | 0°-65°                                             | Monofacial*            | The electricity production of a system is analyzed. According to the obtained results, semi-annual, seasonal and monthly tilt adjustments provide between 0.5 % and 1 %. Tilt adjustments have also additional benefits due to snow sliding.                                                                      | Mitigation | Design:<br>Tilt angle adjustment                     |
| [10] | Norway<br>Germany<br>Switzerland<br>Oslo<br>Tromsø<br>Munich<br>Davos | 10°                                                | Monofacial             | This study found that the mechanisms affecting the net energy balance of PV snow mitigation systems are highly influenced by the climate in which they are installed.                                                                                                                                             | Mitigation |                                                      |
| [63] | Canada<br>Kingston                                                    | 10°<br>20°<br>30°<br>45°<br>55°                    | Monofacial             | The authors have used Darcy's law in this paper to predict the rate of meltwater drainage from the snow cover tilted PV. Experiments have been also conducted to validate the model.                                                                                                                              | Mitigation | Models:<br>Snow melting and sliding prediction model |
| [64] | Romania<br>Brasov                                                     | 47°                                                | N/A                    | In this paper a comparative study has been conducted using infield response of five types of PV modules (p-Si, m-Si, CIGS, CdTe and CIS). The results show that p-Si modules are well-suited if rough glazed PV surface is avoided.                                                                               | Impact     | Design:<br>PV technology assessment                  |
| [13] | Norway<br>Porsgrunn                                                   | 35°                                                | Bifacial               | The performance of bifacial system has been investigated. Annual snow losses have been found to be 0.6 %.                                                                                                                                                                                                         | Impact     |                                                      |
| [32] | Sweden<br>Linköping                                                   | 40°                                                | Monofacial<br>Bifacial | This work presents one year field study in Sweden using bifacial PV compared to monofacial where the snow has contributed to the increase of the bifacial yield up to 48 %. It has been recommended to mount the PV in a higher position to exploit the positive effect of snow in increasing the bifacial yield. | Impact     | Design:<br>Increasing the height of PV panels        |
| [65] | China<br>Jilin<br>Changchun                                           | 27.04°<br>38.08°                                   | Monofacial             | In this work, a novel domino-like snow removal system has been proposed. The process consumes almost no electricity from the grid. The experimental results demonstrate that long-term process does not contribute to the performance degradation of PV cells.                                                    | Mitigation | Mitigation:<br>Self-heating                          |
| [5]  | USA                                                                   | N/A                                                | N/A                    | The authors used the ambient temperature and performance ratio to identify snowfall on PV panels. Results of this approach were compared to actual images of site which have led to 98 % accuracy. This method has reported its effectiveness compared to snow depth data approaches in identifying snow events.  | Impact     |                                                      |
| [38] | Canada<br>Ontario                                                     | N/A                                                | Monofacial             | This work proposes the design, analysis and modelling of a heating system for snow removal from solar PV Electric vehicle (EV) applications. The system relies on a bidirectional DC-DC converter that transfers the grid/EV-battery power into heating of the solar PV modules, leading to snow melting.         | Mitigation | Mitigation: Heating                                  |
| [66] | China<br>Urumqi                                                       | 0°<br>5°<br>10°<br>15°<br>20°<br>25°<br>30°<br>35° | Monofacial             | Numerical simulation has shown that snow accumulation is affected by tilt angle of PV and 1 cm of snow thickness can reduce power generation efficiency to 7.1 %.                                                                                                                                                 | Impact     |                                                      |

(continued on next page)

**Table 1** (continued)

| Ref. | Location                                   | Tilt angle | Technology             | Main findings                                                                                                                                                                                                                                                                                                                        | Category   | Rec.                                                                       |
|------|--------------------------------------------|------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------|
| [37] | China<br><i>Shanghai</i>                   | 40°        | Monofacial             | The paper introduces a novel self-heating snow-removing method for PV panels to address snow effect in cold regions, showing significant improvements in snow-removing time by increasing snow thickness, ambient air temperature, input electric power, and panel tilt.                                                             | Mitigation | Mitigation:<br>Self-heating                                                |
|      |                                            | 45°        |                        |                                                                                                                                                                                                                                                                                                                                      |            |                                                                            |
|      |                                            | 14°        |                        |                                                                                                                                                                                                                                                                                                                                      |            |                                                                            |
|      |                                            | 15°        |                        |                                                                                                                                                                                                                                                                                                                                      |            |                                                                            |
|      |                                            | 16°        |                        |                                                                                                                                                                                                                                                                                                                                      |            |                                                                            |
| [67] | Canada                                     | 17°        | Monofacial             | This paper has investigated the performance of a PV system in snowy conditions and shown that this latter is influenced by module orientation and bypass diodes. The study shown that landscape orientation can reduce the power losses compared to portrait.                                                                        | Impact     | Design: Consideration of module configuration and bypass diodes activation |
|      |                                            | 18°        |                        |                                                                                                                                                                                                                                                                                                                                      |            |                                                                            |
|      |                                            | 30°        |                        |                                                                                                                                                                                                                                                                                                                                      |            |                                                                            |
| [68] | USA<br>Canada                              | N/A        | N/A                    | The paper quantifies the losses due to snow cover in the Northern Western Hemisphere under different climate change scenarios and highlights the substantial reduction in snow losses for PV systems in North America as the climate changes. In many studied areas, snow losses are estimated to be below 5 % in next 20 years.     | Impact     |                                                                            |
| [20] | USA<br><i>New Jersey New York Colorado</i> | Latitude   | N/A                    | This work explores the significance of including snow effect models into PV performance projections using data from the NSRDB TMY2 and TMY3 datasets, highlighting the requirement for reliable historical snow depth data. It also suggests creating a 15-year continuous weather history database to increase simulation accuracy. | Impact     | Incorporating snow depth into datasets                                     |
| [21] | Germany                                    | N/A        | N/A                    | This work describes an empirical approach to improving PV power estimates during snow cover, which has been evaluated and found to minimize forecast errors.                                                                                                                                                                         | Impact     | Models:<br>PV power prediction and forecasting snow cover                  |
| [69] | Norway<br><i>Oslo</i>                      | 10°        | Monofacial             | Heating as snow mitigation has been assessed indoor and outdoor. Economic payback time has been also calculated, showing the importance of the control of peak load due to heating needs.                                                                                                                                            | Mitigation | Mitigation:<br>Use of weather forecasts in mitigation planning             |
| [70] | Switzerland                                | 30°        | N/A                    | In this work, the performance of three PV placement scenarios in urban and mountain environments has been compared. A quantitative assessment of snow effect has shown an increase in yield by 10 % considering its high reflectance.                                                                                                | Impact     | Design:<br>Correct placement for better exploiting a snow-covered ground   |
|      |                                            | 35°        |                        |                                                                                                                                                                                                                                                                                                                                      |            |                                                                            |
|      |                                            | 40°        |                        |                                                                                                                                                                                                                                                                                                                                      |            |                                                                            |
|      |                                            | 65°        |                        |                                                                                                                                                                                                                                                                                                                                      |            |                                                                            |
|      |                                            | 90°        |                        |                                                                                                                                                                                                                                                                                                                                      |            |                                                                            |
| [71] | USA<br><i>Utah</i>                         | 0°         | Bifacial<br>Monofacial | Snow shedding capabilities have been compared for bifacial and monofacial PV panels. The site albedo has increased up to 75 % in presence of snow resulting in heat and electrical generation increase for bifacial PV. This increase in heat contributed to snowmelt on the front side of the panel.                                | Mitigation |                                                                            |
| [22] | USA<br><i>Denver</i>                       | N/A        | N/A                    | This paper proposes a computational intelligence-based short-term PV power forecasting approach that is compatible with snow conditions and independent of exogenous weather forecasts.                                                                                                                                              | Impact     |                                                                            |
| [72] | China<br><i>Xi'an</i>                      | N/A        | N/A                    | A novel snow recognition algorithm for PV modules based on images taken by drones and surveillance cameras is proposed. It analyses the morphology of snow cover as well.                                                                                                                                                            | Impact     |                                                                            |

**Table 2**

Summary of published reviews regarding snow effects on PV systems.

| Ref. | Snow losses estimation | Snow formation | Influencing factors | Snow removal approaches | Forecasting techniques | Year |
|------|------------------------|----------------|---------------------|-------------------------|------------------------|------|
| [2]  |                        |                | ✓                   | ✓                       |                        | 2013 |
| [26] |                        |                |                     | ✓                       |                        | 2017 |
| [73] |                        | ✓              |                     | ✓                       |                        | 2018 |
| [23] | ✓                      |                | ✓                   | ✓                       |                        | 2019 |
| [27] |                        |                |                     | ✓                       |                        | 2020 |
| [28] |                        |                |                     | ✓                       |                        | 2020 |
| [29] |                        | ✓              | ✓                   | ✓                       |                        | 2021 |
| [25] |                        |                |                     |                         | ✓                      | 2022 |

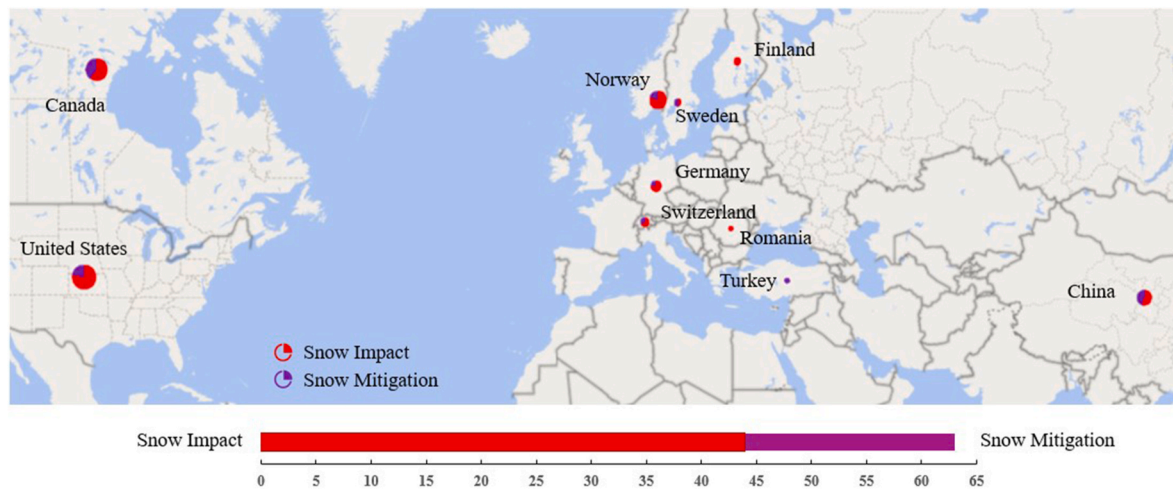
### 3.2. Deposition

Snow particles can be deposited on PV panels through different mechanisms such as direct snowfall and wind-induced deposition because of the transport mode as previously described. The deposition process, in combination with other mechanisms such as irregular melting and the presence of larger-scale roughness elements formed by wind erosion, contributes to the varied roughness scales observed on snow-covered surfaces [75]. It has been reported by Ref. [3] that in the absence of a wet surface, deposition of dry snow particles on a vertically mounted panel will not accumulate due to low friction and cohesion, in contrast to smaller tilt angles. In fact, it should be noted that there is a

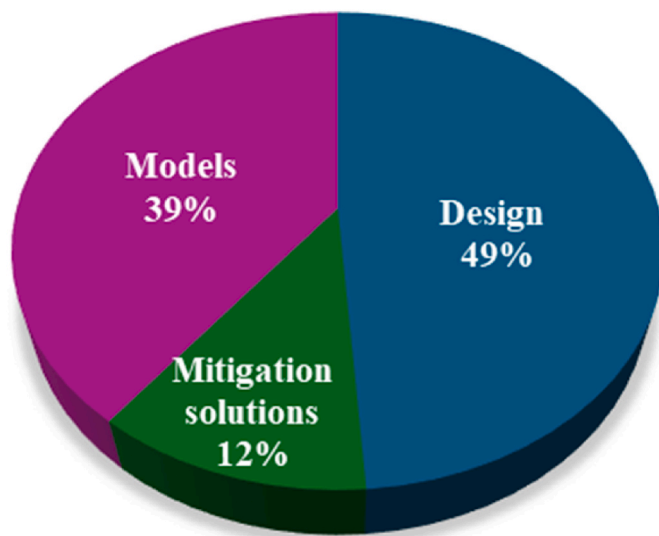
noticeable gap in the state-of-the-art research on snow deposition on PV panels as a process, with most studies concentrating on snow loss effects and snow removal.

### 3.3. Accumulation

The accumulation of snow refers to the gradual continuous increase in snowfall over time forming a snow cover on a given surface. The way snow is spread out across a big area and even on smaller scales, is mostly because of the wind. This interaction between the wind, snowfall, and the ground where the snow lands determine how much snow accumulates and builds up in different places during winter [76]. In a related



**Fig. 1.** Geographical distribution of studies related to snow effect on photovoltaic performance around the world, where the size of the dots corresponds to the number of publications and the bar at the bottom refers the total number of publications on snow impact vs snow mitigation.



**Fig. 2.** Distribution of published works by recommendations as reported in 43 papers.

context, snowdrifts are the result of snow eroding from exposed regions and collecting in sheltered locations, which causes the redistribution of snow due to strong winds [57]. This phenomenon, quantified by the friction velocity, can in polar climates lead to accumulation of large amounts of snow on PV modules, both restricting power generation and putting a mechanical load on the PV array [8]. These authors investigated adjusting solar power plant designs to enhance snowdrift resilience in local climates while minimizing negative impacts on energy yield. The authors found that increasing the gap-to-ground and system scale were effective adjustments for reducing snowdrift accumulation risk without significantly affecting energy production. In a related study [60], they concluded that adapting solar power plant designs is essential to mitigate snowdrift development and optimize system performance in such climates.

The authors [29] reviewed the effects of parameters such as the mass concentration of snow particles in the air, precipitation rate, and wind speed on snow accumulation and surface texture on for various systems. In the following section, we will address the impact of snow accumulation on PV systems in more detail, highlighting the most influential factors impacting snow losses.

#### 4. Factors influencing snow losses

Various factors contribute to the deposition, generation of losses and natural clearing of snow. These factors have been classified into four categories, as illustrated in Fig. 4: Surface properties, installation characteristics, snow properties, and climatic conditions. Surface properties include characteristics like hydrophobicity, hydrophilicity, and roughness. Installation characteristics include parameters like tilt angle, orientation, PV technology, gap to ground, and pitch. Snow properties include thermal and mechanical properties. Climatic conditions comprise snowfall, ambient temperature, wind speed, and solar irradiance. Understanding and addressing these diverse factors are essential for mitigating snow-related losses and optimizing the performance of PV systems in snowy environments.

##### 4.1. Installation characteristics

###### 4.1.1. Tilt angle

Studies have explored optimizing solar panel tilt angles to minimize snow accumulation and maximize energy production, particularly in high-latitude regions. Authors of [73] have found that tilting panels upwards reduces snow buildup. It was found by Ref. [60] that higher tilt angle of around  $45^\circ$  combined with snow removal geometries like the gap to ground are recommended to maintain the initial energy production. In a previous study [47], it was found that high tilt angles can increase PV yield considering the effect of increasing albedo. A lower tilt angle increases the risk of snow accumulation and reduces energy output significantly [66] as illustrated in Fig. 5. Additionally, there is evidence that snow sliding is negligible at tilt angles smaller than  $10^\circ$  [11]. It was suggested by Ref. [8] using variable tilt systems or adjusting tilt angles seasonally, though this may incur additional costs [31]. proposed an optimal tilt angle for snow impact, offering an  $8.5^\circ$  improvement over traditional angles but requiring further economic analysis. However, high tilt angles will often not be optimal with respect to maximizing irradiance through the year or maximizing ground/roof coverage [55] and reducing self-shading. Overall, balancing snow mitigation and energy production through strategic tilt angles remains an important consideration for solar power plants in cold climates.

###### 4.1.2. Pitch

Pitch refers to the distance between rows of PV panels, which is carefully chosen to achieve the desired ground coverage ratio (GCR) and minimize shading [77]. In the work of [8] it is shown that a reduced pitch increases the friction velocity, previously defined, which reduces

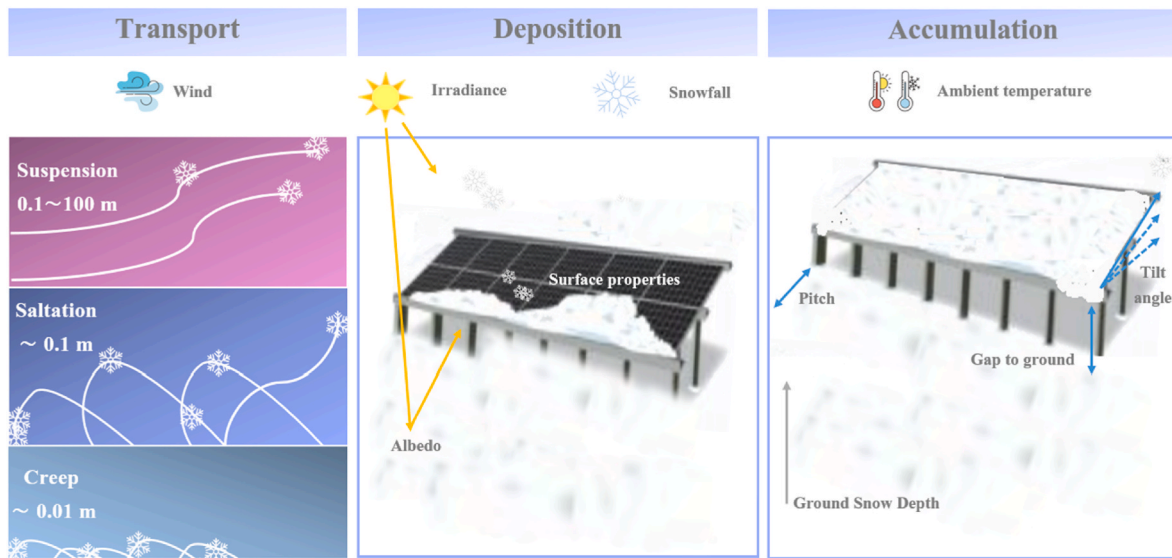


Fig. 3. Illustration of snow soiling process involving transport, deposition and accumulation steps.

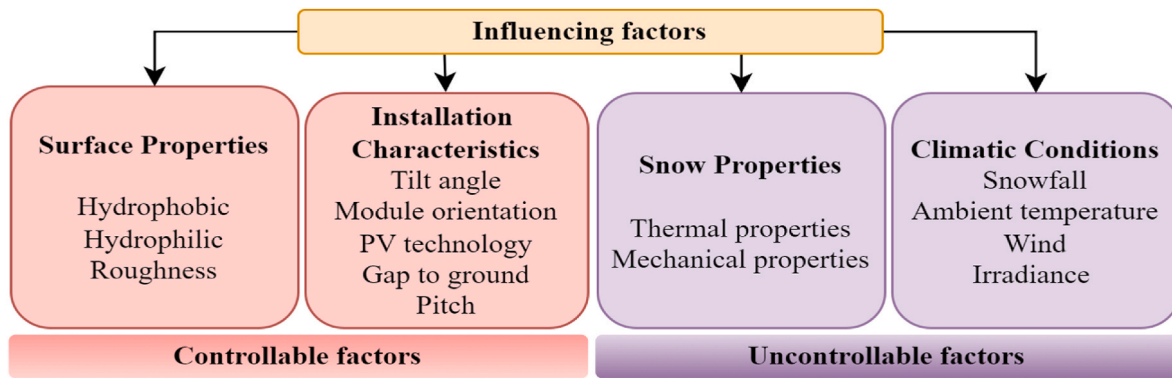


Fig. 4. Factors influencing snow losses.

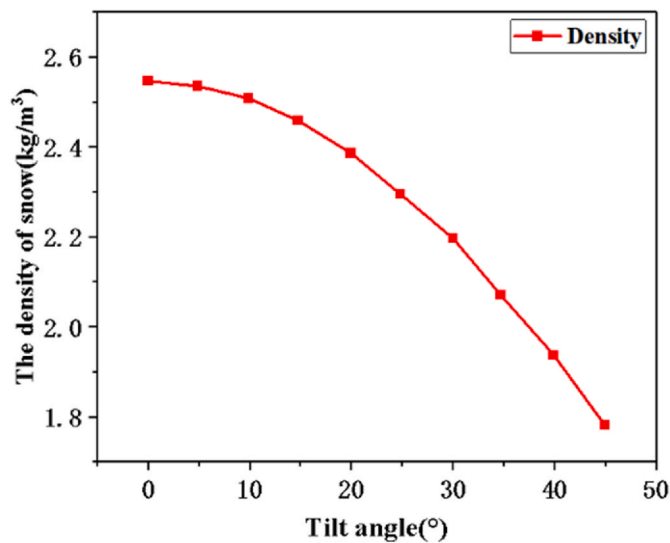


Fig. 5. Relationship between tilt angle and snow density on the surface of PV modules [66].

the risk of snowdrift accumulation. However, this study did not investigate the snowdrift accumulation effect on snow losses of PV plants. In the same context [42], found that increased pitch can accelerate snow clearance from both the panels and the surrounding area throughout the late winter and early spring seasons due to less interrow shading.

#### 4.1.3. Gap to ground

The work of [8] found that increasing the gap between the panels and the ground (gap-to-ground) is a mitigation strategy of snowdrift accumulation with minimal impact on energy. Snow accumulates on the ground under the solar panels due to turbulence behind them. As a result, in practical engineering, it is feasible to elevate the panel's elevation above the ground surface to enable faster snow sliding [66]. According to the study of [60], three raised modules had snow-related energy losses between 5 % and 12 %, while similarly inclined modules placed adjacent to the ground experienced losses between 29 % and 34 %. As a recommendation of [32], mounting the PV in a higher position to exploit the positive effect of snow albedo is useful to increase the bifacial yield.

Snow guards are frequently installed on rooftops in areas with heavy snowfall to enhance safety by preventing snow from sliding off and potentially injuring people [27]. These safety devices slow down snow movement, reducing its shedding. Therefore, their use on roofs with an optimized PV design featuring an increased gap to ground should be carefully evaluated. A gap has been identified in literature regarding the use of snow guards in the presence of PV modules.

#### 4.1.4. Module orientation

There has been literature on the effect of module orientation (portrait or landscape) on snow shedding and subsequent power loss. Lower energy losses during partial snow cover are reported when the landscape orientation is used, due to the activation of bypass diodes [60, 67]. Preliminary results from image analysis of one snow shedding event indicate that PV modules installed in portrait orientation yield up to 24 % more energy yield compared to those in landscape orientation due to faster snow shedding [35].

#### 4.1.5. PV technology

Several studies show that systems with bifacial modules have lower snow losses than systems with monofacial modules [4]. found monofacial snow losses of in average 33 % for winter period, and 16 % on an annual basis for a site in Escanaba. In this study, they also found that bifacial systems perform better than monofacial in severe winter conditions, and that the annual bifacial losses were 2 % in the worst-case scenario. Another comparison has been conducted by Ref. [32] based on one year field study using bifacial PV compared to monofacial. In this study, the snow contributed to an increase of the bifacial yield up to 48 %. For BAPV bifacial system, snow losses were found to be 0.6 % [32]. In a temperate continental climate with snowy winters, the results of [64] experienced the lowest relative efficiency loss. The study suggests that p-Si modules are well-suited compared to m-Si and CIGS, as long as rough glazed PV surfaces are avoided. The findings of another work [54] show that thin film modules, ideally unframed, can become a very interesting option in snowy days as they exhibit higher snow shedding rates compared to framed modules. The advantages of frameless PV were also reported by Ref. [34] in a different location in USA.

### 4.2. Climatic conditions

#### 4.2.1. Snowfall

Snowfall is the amount of snow that falls in a determined period. It is generally daily recorded as liquid water equivalent when snow occurs [78]. It refers to the direct mechanism of snow deposition as illustrated in Fig. 3. The authors [23] have classified the published works into three categories based on ambient temperature and the amount of snowfall: mild, modest, and severe climates. The corresponding loss in electricity generation due to snow was also reported. For instance, in severe climates characterized by winter temperatures consistently below 0 °C and significant snowfall [61], found that snow losses averaged between 6 % and 26 % for an annual snowfall of 711 cm in San Ramon, California. In another study [31], it was found that annual snowfall at the study site, located near Stockholm, can reach 50 cm or more, with corresponding annual electricity generation losses of 14.7 %. In Prince Edward Island in Canada where the annual snowfall was 186 cm, the work of [41] has reported 2.7 % in energy loss due to snow. In USA [79], have found respectively annual energy loss of 16 %, 2.9 % and 2 % in Escanaba, Colorado, and North Carolina where annual snowfall were around 124 cm, 137 cm and 25 cm. In Norway [13], has found 0.6 % annual losses due to snow in Porsgrunn where annual snowfall was around 98 cm. It should be noted that most of the cited works do not mention the recorded annual snowfall and data was obtained from Refs. [80,81].

Annual snowfall patterns are changing in response to climate change [68]. These authors evaluated snow-related losses in potential PV electricity generation across various regions of the Northern Western Hemisphere. Their findings indicate that snow losses are expected to remain below 5 % over the next 20 years. However, further research is required to assess the broader impacts of climate change on PV performance, including operating temperatures and snow-related losses.

#### 4.2.2. Wind

Wind is the main factor responsible for the transport of snow particles in the entire process as illustrated in Fig. 3. Strong wind and little shelter cause significant horizontal redistribution of snow, which leads

to development of snowdrifts in the aerodynamic shade of the PV arrays in polar climates [57]. The development of the snowdrift can cover the PV plant, thus significantly impacting the power production. Indeed, the interference of wind with tilt angle has been investigated in recent work of the same author [8]. It has been shown that wind speed in the system can be altered by raising the tilt angle, resulting in discrete accumulation of snow and erosion zones. Regarding wind direction, the same study has found that prevailing wind direction can significantly increase the friction velocity if it comes closer to parallel to the PV rows.

It has been found by Ref. [36] that wind has a significant effect on snow melting. In this case, a wind of an average velocity of approximately 2 m/s at around 3 °C air temperature, has increased snow melting by 90 %. Wind speed has an important effect on snow sticking efficiency, which is defined as the ratio of the number of particles that attach to the object to the number that collide with it [29]. According to these authors, with higher velocity, snow particles can more effectively re-bounce by overcoming the interfacial forces between them and substrate. As a result, the sticking efficiency decreases with wind speed.

#### 4.2.3. Ambient temperature

The effect of ambient temperature is mostly linked to snow sliding or melting and icing. It was demonstrated by Ref. [11] that both irradiance and increasing air temperature can increase the snow temperature to melting. In general, the temperature of the ambient air determines heat convection between the snow-covered panel and the surrounding air, as well as heat radiation from the snow's surface to the sky [37]. The same work has predicted the snow removing duration with the increase of 1 °C in ambient temperature based on a simplified model that has been validated with experimental data. In this context, the authors [5] found that snow accumulation on PV arrays is more accurately identified by relying on ambient temperature rather than external snow depth data. It has been reported in Ref. [54] that times of initial sliding correspond to times at which either the ambient temperature or the irradiance rise, which will most likely lead to an increase in the module temperature. The ambient temperature is also considered an important factor in sticking efficiency of snow particles. It has been reported in Ref. [36] that the sticking efficiency increases with temperature based on the results of [82]. It should be noted that all prediction models of the effect of snow on PV are significantly dependent on the ambient temperature [23].

#### 4.2.4. Irradiance

Snow cover is directly influenced by solar irradiance. Higher irradiance levels received by PV panels can lead to increased warming, making it easier for snow to melt and slide off, ultimately resulting in its removal [23]. Indeed, irradiance is considered the most significant input in snow losses estimation techniques in particular [83] and PV power prediction models [25]. Table 4 presents a summary of models where irradiance is one of the used inputs. More detailed studies regarding the impact of irradiance have been recently published. It has been found that the front absorbed irradiance significantly improves the prediction model compared to those based on plane of array irradiance [44].

### 4.3. Surface properties

Surface roughness of PV has been studied in literature [64] as it can increase snow retention and restrict irradiance access. Hydrodynamic coatings have been investigated in the past to increase ice clearance on glass surfaces [39] by lowering adhesion and friction, or by absorbing the solar irradiance to break down the snow cover [23]. Hydrophobic, superhydrophobic, and hydrophilic coatings are used to passively reduce snow on PV modules [2,29] and dust accumulation [84]. It was reported that a super-hydrophobic film has demonstrated remarkable anti-snow adhesion properties against both dry and wet snow, as well as sliding acceleration with dry snow. However, wet snow exhibited poor sliding behavior on the film, and the hydrophilic smooth glass surface

**Table 3**

PV energy production losses caused by snow. It should be noted that the climate was not mentioned in most of the works and the classification has been done according to Koppen Geiger based on the location of the study.

| Ref. | Region <sup>a</sup>                  | Climate                        | Technology             | Energy production loss | Frequency            |
|------|--------------------------------------|--------------------------------|------------------------|------------------------|----------------------|
| [31] | Sweden<br>Hammarby<br>Sjöstad        | Dfb                            | Monofacial             | 14.7 %                 | Annual               |
| [66] | China<br>Xinjiang                    | BWk                            | Monofacial             | 7.1 % <sup>b</sup>     | Dependent occurrence |
| [10] | Norway<br>Tromsø                     | Dfc                            | Monofacial             | >80 %                  | Monthly              |
|      | Switzerland<br>Davos                 | Dfc                            |                        | >80 %                  | Monthly              |
|      | Germany<br>Munich                    | Cfb                            |                        | <22 %                  | Monthly              |
|      | Norway<br>Oslo                       | Dfb                            |                        | >50 %                  | Monthly              |
| [41] | Canada<br>Prince<br>Edward<br>Island | Dfb                            | Monofacial             | 2.7 %                  | Annual               |
| [11] | USA,<br>Colorado<br>Wisconsin        | Bsk<br>Dfb                     | Monofacial             | 1 %–12 %<br>90 %       | Annual<br>Monthly    |
| [55] | Norway<br>Kjeller                    | Dfb                            | Monofacial             | Up to 100 %            | Monthly              |
| [14] | Northern<br>Sweden                   | Dfc                            | Monofacial             | 20 %                   | Annual               |
| [43] | USA                                  | Temperate<br>zone <sup>a</sup> | Monofacial             | 54.5 %                 | Monthly              |
| [4]  | USA<br>Escanaba                      | Dfb                            | Monofacial<br>Bifacial | 16 %<br>2 %            | Annual               |
| [19] | USA<br>Colorado                      | Bsk                            | Monofacial             | 2.9 %<br>~ 12 %        | Annual<br>Monthly    |
| [7]  | USA<br>North<br>Carolina             | Cfa/Cfb                        | Monofacial             | 2 %                    | Annual               |
| [56] | Canada<br>Cambridge<br>Bay<br>Ottawa | ET<br>Dfb                      | Bifacial               | 0.034 %                | Daily                |
| [13] | Norway<br>Porsgrunn                  | Cfb                            | Bifacial               | 0.6 %                  | Annual               |

<sup>a</sup> The climate has been specified by the authors for most sites using PV climate zones classification.

<sup>b</sup> for 1 cm snow thickness.

was thus only effective for sliding wet snow [85]. Earlier [39] found that hydrodynamic coatings are not very efficient regarding snow removal. However, using coatings with reduced snow adhesion can be of great added value to maximize power generation from PV modules during the winter period [30].

#### 4.4. Snow properties

The variability of snow particles and snow cover reflects the well-known complexity of the mechanisms involved in the formation of snow and, consequently, snow covers [3]. Snow is characterized by various mechanical properties including adhesion, cohesion, tensile strength, shear modulus and others as well as thermal properties as detailed in the review of [29]. These properties are crucial in predicting the breakup as well as the movement of the snow layers. In this work, we will not address these properties as they have been well discussed previously but a focus will be given to adhesion as an important property that inhibits the snow shedding, albedo and thermal conductivity [36].

##### 4.4.1. Adhesion

Adhesion is the tendency of dissimilar materials to stick to each other. In the context of this paper, it is manifested by the formation of ice

on the PV panel surface [3]. It is highly impacted by the LWC and the surface properties on which snow is accumulated. The LWC, or wetness of snow, is defined as the ratio of the mass of water contained to the total mass of the wet snow [86]. Depending on LWC, snow is commonly separated into two types: wet and dry snow. While dry snow corresponds to temperatures below  $-1^{\circ}\text{C}$  to  $-2^{\circ}\text{C}$  [87], wet snow is associated with temperatures that are near freezing [27]. In the work of [88], a methodical approach has been developed to assessing how surface energy and roughness affect snow adhesion and its shedding mechanism. These authors found that snow adhesion increases with the increase of LWC and strongly adheres to the surfaces when the LWC is around 20 %. Once this value surpassed, snow adhesion decreases. Results regarding surface roughness have also been obtained. It was reported in the same study that on a smooth surface, snow has experienced low adhesion strength. At intermediate surface roughness, the contact between snow and substrate increases which increases its adhesion strength. However, data regarding snow adhesion is limited [89].

##### 4.4.2. Albedo

For PV yield, the most important and frequently measured optical property of snow is its albedo [90]. The albedo and transmittance are regarded as apparent optical characteristics. The albedo of snow is generally known in literature by its positive effect in increasing the PV yield [70]. The increase in yield is particularly obtained in the case of high tilt angle systems [47]. According to Ref. [4] the impact of snow albedo relates to improved performance of bifacial PV, which can outperform monofacial PV in harsh winter conditions. In the same context, the ground albedo has been reported to increase up to 75 % in presence of snow, resulting in heat and electrical generation increase for bifacial PV. This increase in heat contributed to snowmelt on the front side of the panel [71].

##### 4.4.3. Thermal conductivity

The thermal conductivity of snow refers to its ability to conduct heat. Snow is considered a good heat insulator [29] characterized by low thermal conductivity ranging between 0.04 W/mK [47] and 0.09 W/mK depending on snow density [26]. According to literature, it is rarely studied *sensu stricto* because heat transfer occurs not only by conduction through the interconnected snow grains but also through latent heat transfer and heat conduction in air in the pores [91]. Thus, most of studies [3,63,92,93] have assessed or referred to the effective thermal conductivity of snow and defined it as a function of heat flux and temperature gradient. It has been reported that the effective thermal conductivity increases as temperature decreases [93]. As previously mentioned, the LWC affects several properties of snow, and the thermal conductivity increases with the increase of the LWC of snow [29].

Fig. 6 summarizes the factors involved in PV system snow losses, extracted from previous studies that explored each factor independently as well as their interactions. In this work, the significance of these effects has been evaluated and classified as low, moderate, or significant. Specific combinations of these factors can lead to large variations in losses.

#### 5. Snow losses

The accumulation of snow on solar panels significantly reduces their performance, gradually leading to a complete block of irradiance and a total loss of production. According to the study of [90], it has been demonstrated how transmittance rapidly decreases as snow depth rises, and this is especially true for the visible wavelength range as shown in Fig. 7, which [73] has redone.

Table 3 highlights the electricity generation loss caused by snow cover in past studies as reported by various authors and can be considered as a continuation of the works by Ref. [23]. The study conducted by Ref. [11] has been mentioned in this table as a way of reporting both annual and monthly snow losses for the same location.

**Table 4**

A summary of models up to now with the inputs, output and their limitations. \*The output is mentioned as reported in the published works.

| Models                       | Powers et al. [15]                                         | Lorenz et al. [21]                                   | Andrews & Pearce [17]                                                     | Marion et al.; Pisklak;<br>Severin Ryberg & Freeman<br>[11,20,83] | Shishavan et al. [16]                                                     | Bosman & Darling [24]                                                         | Hashemi et al. [18,<br>19]                            | Hashemi et al. [22]                                                   |
|------------------------------|------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Inputs</b>                |                                                            |                                                      |                                                                           |                                                                   |                                                                           |                                                                               |                                                       |                                                                       |
| Tilt angle                   | ✓                                                          |                                                      |                                                                           | ✓                                                                 | ✓                                                                         |                                                                               |                                                       |                                                                       |
| Snowfall                     | ✓                                                          |                                                      | ✓                                                                         |                                                                   |                                                                           |                                                                               | ✓                                                     | ✓                                                                     |
| Snow depth                   |                                                            | ✓                                                    |                                                                           |                                                                   |                                                                           |                                                                               |                                                       |                                                                       |
| Ground snow<br>depth         |                                                            |                                                      |                                                                           |                                                                   |                                                                           | ✓                                                                             |                                                       | ✓                                                                     |
| Snow amount                  |                                                            |                                                      |                                                                           | ✓                                                                 |                                                                           |                                                                               |                                                       |                                                                       |
| Days of<br>snowfall          |                                                            |                                                      |                                                                           |                                                                   | ✓                                                                         | ✓                                                                             |                                                       |                                                                       |
| Date time                    |                                                            |                                                      |                                                                           |                                                                   | ✓                                                                         | ✓                                                                             |                                                       | ✓                                                                     |
| Ambient<br>temperature       |                                                            | ✓                                                    | ✓                                                                         | ✓                                                                 |                                                                           | ✓                                                                             | ✓                                                     | ✓                                                                     |
| Module<br>temperature        |                                                            |                                                      |                                                                           |                                                                   |                                                                           |                                                                               |                                                       | ✓                                                                     |
| PV power                     |                                                            | ✓                                                    |                                                                           |                                                                   |                                                                           |                                                                               |                                                       | ✓                                                                     |
| Plane of Array<br>Irradiance |                                                            |                                                      | ✓                                                                         | ✓                                                                 |                                                                           |                                                                               | ✓                                                     | ✓                                                                     |
| Sliding<br>Coefficient       |                                                            |                                                      |                                                                           | ✓                                                                 |                                                                           |                                                                               |                                                       |                                                                       |
| Humidity                     |                                                            |                                                      |                                                                           |                                                                   |                                                                           |                                                                               | ✓                                                     |                                                                       |
| Wind speed                   |                                                            |                                                      |                                                                           |                                                                   |                                                                           |                                                                               | ✓                                                     |                                                                       |
| Wind direction               |                                                            |                                                      |                                                                           |                                                                   |                                                                           |                                                                               | ✓                                                     |                                                                       |
| Air pressure                 |                                                            |                                                      |                                                                           |                                                                   |                                                                           |                                                                               |                                                       | ✓                                                                     |
| <b>Output*</b>               | Monthly energy<br>production                               | Daily<br>PV power                                    | Daily<br>PV power and current                                             | Monthly and annual<br>energy production                           | Monthly snow loss                                                         | Monthly energy<br>production                                                  | Monthly and annual<br>snow loss                       | Hourly PV power                                                       |
| <b>Limitation(s)</b>         | Limited inclusion of<br>influential factors in<br>the test | Fog events<br>challenge weather<br>prediction models | Limited open data sharing<br>restricts the generalization<br>of the model | Difficulty of<br>incorporating snow<br>depth into datasets        | Validation of estimated<br>loss factors is needed for<br>greater accuracy | Inability to use TMY3 data<br>due to the absence of snow<br>depth information | Limited applicability<br>to small, new PV<br>datasets | Higher errors in<br>forecasting hours ahead<br>considering snow cover |

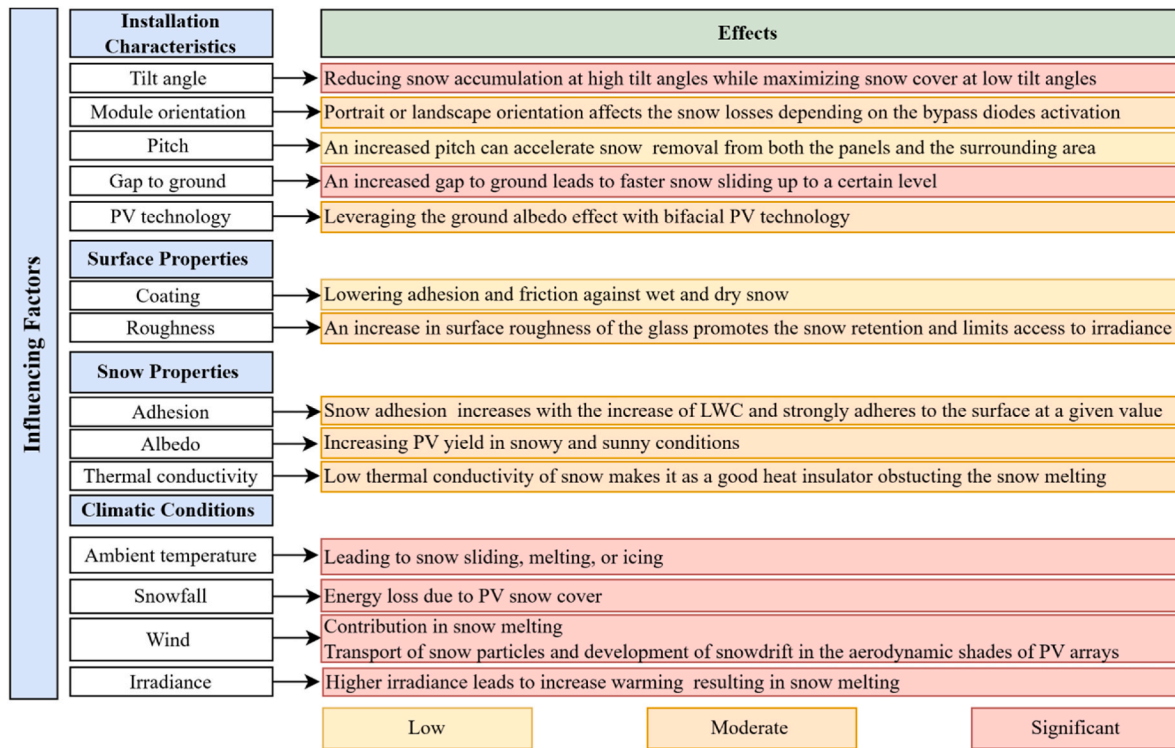


Fig. 6. Summary of factors involved in PV system snow losses with their corresponding effects.

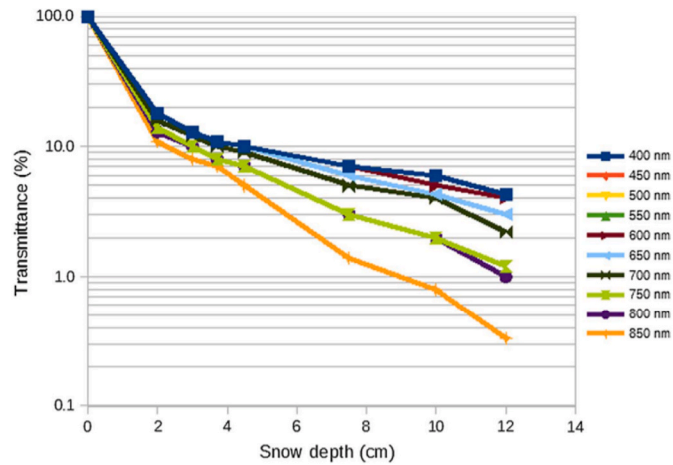


Fig. 7. Transmittance measurements vs snow depth as carried out by Ref. [90] and redone by [73].

The highest monthly snow losses have been reported between 80 % [10], 90 % [11] and 100 % [55]. Annual snow losses are characterized by low magnitude compared to monthly. In Ref. [13], reported snow losses were 0.6 % for a given year, whereas [14] found annual losses up to 20 %. The reported losses in energy due to snow are presented in Fig. 8 per technology as well as Koppen Geiger climate zone. Among conducted studies, it was found that 16 % have been estimated for monofacial panels compared to 2 % for bifacial PV [4,79]. Studies dealing with snow impact on trackers are rare [56]. has reported a daily energy yield loss of 0.034 % per centimeter of accumulated snow for single axis tracker compared to 7.1 % for fixed PV systems [66].

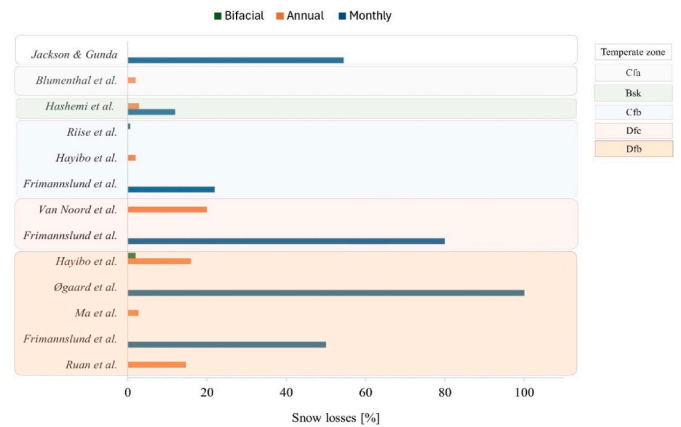


Fig. 8. Reported losses in energy due to snow in studies per technology and Köppen Geiger Climate zone. Bifacial is given in annual frequency while monofacial technology is given by both monthly and annual frequency.

## 5.1. Snow losses estimation approaches

### 5.1.1. Snow losses modelling

Besides models that can provide direct energy loss considering snow effect [15–17], other models can be categorized as snow cover prediction models as explained previously, which estimate the snow-covered area of a panel [11,20–22]. This information can be used to calculate the generation loss. Threshold-type and first principle thermal heat and mass transfer models have been used to assess snow coverage on PV panels as reported by Ref. [23]. As first principle models need more validation studies, in this paper a focus will be given to threshold-type models as they provided the best predictions utilizing easily available data from weather stations up to now. To the best of the authors' knowledge, the limitations of some existing models have not been explicitly addressed in recent reviews. Consequently, this work aims to

shed light on these unexplored limitations that require attention. In Table 4, a summary of these models is presented while precisizing the inputs of the model, its output, and the reported limitations up to now.

As shown in section 4, in PV plants, snow losses are very challenging as the effect depends not only on the local environment but also on further complex factors like the amount and form of the snow layer on each individual panel, the tilt angle of the arrays, albedo radiation and other factors. This is why some works have attempted to address this complexity by utilizing machine learning techniques to formulate a prediction problem [25]. [22] has proposed a computational intelligence-based short-term PV power forecasting approach that is compatible with snow conditions and independent of exogenous weather forecasts. However, these authors have also raised concerns regarding the increase of forecasting errors for hours ahead depending on snow cover conditions.

### 5.1.2. Image analysis

Image analysis as a critical method for snow losses detection in PV systems has been highlighted by recent studies. In Ref. [47], extracted performance data have been used to validate a novel methodology based on image analysis to determine the probability distribution of snow deposition on PV modules. The importance of system design in snow mitigation measures was underlined by Ref. [35] who used image analysis to show that PV modules in portrait orientation can generate up to 24 % more energy than those in landscape orientation during snow shedding events. Similar to this [79], analyzed snow losses on various PV systems using open-source image processing techniques and empirical data. The study highlights the importance of utilizing image-based techniques to understand the impact of snow accumulation on energy generation in northern environments. Furthermore [52], demonstrated advances in using image analysis for precise snow assessment by proposing a novel method combining image processing and deep learning techniques, such as LMB and CNN, to quantify snow impact on PV panels. Moreover [54], used image processing to investigate the behavior of snow melting on various PV technologies. They found that large frameless modules had significant snow removal rates and highlighted the usefulness of image analysis in comprehending the dynamics of snow management for maximizing PV system performance.

Regarding the limitations, several inaccuracies in snow covering can occur based on image analysis. PV modules' characteristics, colors, smoothness, and contrast change significantly before and after ice and snow buildup, resulting in distinct morphological aspects [72]. For example, the frame can be lightly colored, thus classified as snowy area which underestimates the modelled power. To decrease these errors, the angle of the camera can be chosen near normal to the face of the PV module [35]. Thus, in the work of [72] drones and surveillance cameras have been used to collect comprehensive image data to determine the extent of snow and ice coverage.

## 5.2. Snow removal approaches

### 5.2.1. Heating

Heating is one of the proposed active snow mitigation solutions [29]. Two heating methods were investigated in the work of [33], including electrical heating by a resistance thin-film heater mounted on the back of the PV modules and by the reverse current through PV cells. In another work of the same author [46], circulating hot fluid through the back of the panel has been proposed for removing snow from PV/T panels. This method has shown its effectiveness to clear the PV panel. Nevertheless, the energy required to warm a panel to the point of clearing in a cold region could exceed the energy generated because of the clearing [23]. Self-heating for PV as a snow removal method involves using the heat generated by the panels themselves to melt snow. This is achieved by applying a certain positive voltage to the panels, causing them to act as a load and generate heat [37,59]. The process of snow melting involves a lower layer which consists of wet snow

saturated with water, while the upper layer remains dry. The wet snow layer increases in thickness due to the combined action of capillary forces and gravity until it achieves its maximum height (also known as the equivalent height), which is stated in centimeters for a specific snow density [59]. It was reported that self-heating can effectively remove snow from PV panels when the snow thickness exceeds the equivalent height and the tilt angle of the PV panels is greater than the required minimum tilt angle, as determined by a mechanical model that relies on the surface friction coefficient [59]. Significant improvements in snow-removing time were found by increasing the factors of snow thickness, ambient air temperature, input electric power, and panel tilt.

In other applications related to PV electric vehicle [38], proposed the design, analysis and modelling of a heating system that relies on a bidirectional DC-DC converter that transfers the grid/EV-battery power into heating of the solar PV modules, leading to removing snow from the panel. A novel solution has been proposed by Ref. [65] based on a domino-like snow removal approach. It consists of removing snow off PV systems string by string using the electricity generated from exposed PV modules. The PV string that removes the snow first will supply power for the PV string that follows that is covered in snow. Like the mechanism of dominoes, the total power of "electricity providers" increases step by step as the number of PV strings that have finished removing snow increases. The proposed solution has shown high snow removal efficiency with very low power consumption compared to other methods.

### 5.2.2. Surface coating

Coatings are being attempted to mitigate snow accumulation on PV panels surface and considered as passive solutions since supplying energy is not required [2,23,26–29,73]. Despite the previous discussion on hydrophobic, superhydrophobic, and hydrophilic coatings, icephobic coatings may offer a repellent surface solution against snow [94]. These authors numerically investigated the efficiency of PV cells treated with icephobic coatings available on the market. The study concluded that this type of coating is highly recommended in a continental climate to maximize PV power generation. Additionally [27], noted that snow and ice adhesion should be treated separately from a surface design perspective. In the work of [30], it was found that coatings may be ineffective during snow melting or prolonged melting. This adds to the general issue of durability, which may limit the long-term efficiency of these surfaces [29], as well as the decrease in transmittance resulting from surface treatment [27].

### 5.2.3. Mechanical and natural snow removal

The use of mechanical snow clearing is not well discussed in literature as pointed out by Ref. [23]. Rectangular roof rakes have been tested in the work of [95], showing some challenges regarding shedding snow specially when it is very icy and adhered to the surface. Thus, removing snow completely has been proved to be challenging. In order to avoid scratching the glass surface or the formation of cracks, it was recommended to leave a thin snow cover that will naturally melt during sunny days. In this context [30], has categorized different mechanisms of natural snow removal using a novel sensor that measures the LWC and temperature of the snow at its interface with the modules. Melting, prolonged melting, melting followed by shedding, and shedding (fast sliding) were identified as snow removal. The paper presents the impact of snow thickness as well as the LWC on occurrence and duration of each mechanism.

### 5.2.4. Combination of active and passive solutions

According to what has been previously stated, each snow mitigation solution presents advantages but also limitations as illustrated in Fig. 9. In Ref. [27], it was recommended combining heating and surface treatments, such as superhydrophobicity, that could increase the sliding of snow and ice, allowing the heating energy to be decreased. As natural snow removal by melting or prolonged melting presents significant challenges during winter thus the application of local heating can be

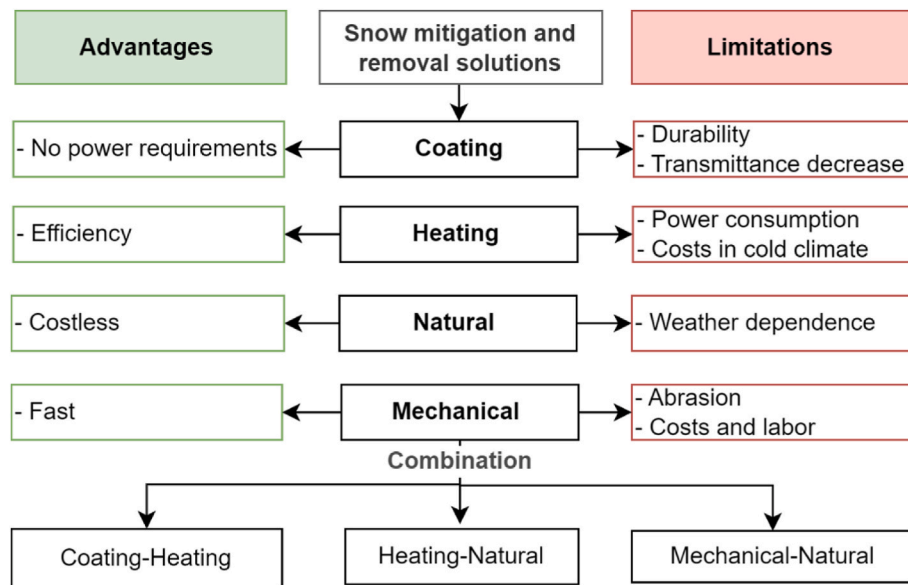


Fig. 9. Advantages and limitations of snow mitigation and removal solutions, along with the recommended combinations of these methods as stated in the literature.

employed to accelerate the snow melting rate [30]. Along similar lines, the combination of mechanical snow removal solutions performed by using rook rakes and melting as previously stated was recommended [95].

## 6. Conclusion

The advancement of PV technology must account for the snowy conditions in countries with severe winters, which means a shift from relying on universal configurations to optimized PV design for harsh winter conditions. Based on the conducted analysis, the key findings of this review are as follows:

- The physics of snow transport, deposition and accumulation are complex where wind has been identified present in the entire process leading to different impact.
- The factors involved in snow losses have been classified into controllable and uncontrollable factors. The number of influential factors is manifested according to the literature review by large variations in reported snow losses.
- One of the recommendations of this review is to enhance the PV system design to make it more suitable for snowy conditions, rather than relying on optimal configurations that neglect the specific challenges of severe winters. These new configurations should consider an optimization of several parameters such as tilt angle, pitch, gap to ground, PV technology and surface properties.
- Further research is needed to address the optimal PV design with the application of safety devices, such as snow guards that limit snow sliding.
- Considering the transition towards bifacial technologies, further studies are needed to evaluate losses due to snow for both bifacial and tracking technologies as well as highlighting the potential gain due to the high albedo of snow.
- Accurate snow loss modeling and shedding prediction require the inclusion of additional factors that impact snow accumulation and clearing, such liquid water content.
- The applicability of existing models can be enhanced through image analysis to avoid inaccuracies in snow covering for diverse PV configurations and locations. This can be achieved by integrating drones, which provide good control of the camera angle and thus improve snow detection.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

No data was used for the research described in the article.

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